University of the Philippines - Baguio

Archie Junio's Reference Notebook

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Contents

1. Data Structures 1
1.1. Segment Tree 1
1.2. Union Find 1
1.3. 2D Sparse Table 1
2. Dynamic Programming 1
2.1. Lichao Tree 1
2.2. Divide And Conquer 2
3. Geometry 2
3.1. Angles And Rotations 2
3.2. Distances 2
3.3. Intersections 2
3.3.1. Circle Line Intersection 2
3.3.2. Circle Circle Intersection 2
3.3.3. Line Intersect 2
3.4. Closest Pair 2
3.5. Convex Hull 2
3.5.1. Convex Hull 2
3.5.2. Convex Hull 3D 2
3.6. Areas 3
3.6.1. Triangle Area 3
3.6.2. Polygon Area 3
3.7. Dot Cross 3
3.8. Line Sweep Closest Pair 3
4. Graphs 3
4.1. Max Flow 3
4.1.1. Dinic 3
4.1.2. Edmonds Karp 3
4.1.3. Bipartite Matching 3
4.2. Shortest Path 4
4.2.1. Dijkstra 4
4.2.2. Bellman Ford 4
4.3. Minimum Spanning Tree 4
4.3.1. Kruskal Min Spanning Tree 4
4.3.2. Prim Min Spanning Tree 4
4.4. Least Common Ancestor 4
4.4.1. Binary Lifting Lca 4
4.4.2. Euler Lca 5
4.4.3. Tarjan Lca 5
4.5. Minimum Mean Weight Cycle 5
4.6. Strongly Connected Components 5
4.7. Heavy Light Decomposition 5
4.8. Euler Path/Cycle 6
4.8.1. Directed Euler 6
4.8.2. Undirected Euler 6
4.9. Tree Isomorphism 7
4.10. Centroid Decomposition 7
5. Math I Algebra 7
6. Math II Combinatorics 7
6.0.1. Lucas Theorem 7
6.0.2. Catalan Numbers 8
6.0.3. Kth Permutation 8
6.0.4. Factoradics 8
7. Math III Number Theory 8
7.0.1. Gcd Subset Counter 8
7.0.2. Linear Diophantine 8
7.0.3. Chinese Remainder Theorem 8
7.0.4. Euler Totient 8
7.0.5. Extended Euclidean Algorithm 8
7.0.6. Modular Exponentiation 8
7.0.7. Mod Inverse 8
7.0.8. Modulo Solver 9
7.0.9. Primitive Root 9
7.0.10. Josephus 9
8. Math IV Numerical Methods 9
8.0.1. Fast Square Testing 9
8.0.2. Simpson Integration 9
9. Strings 9
9.0.1. Longest Common Prefix 9
9.0.2. Hasher 9
9.0.3. Z Algorithm 9
9.0.4. Aho Corasick Trie 9
9.0.5. Knuth Morris Pratt 10
9.0.6. Trie 10
9.0.7. Suffix Array 10
9.0.8. Booth Min String Rotation 11
10. Other Algorithms 11
10.0.1. Longest Increasing Subsequence 11
10.0.2. Cycle Find 11
10.0.3. 2Sat 11
10.0.4. Dpll 11
10.0.5. Stable Marriage 12
10.0.6. Dates 12
10.0.7. Simulated Annealing 12
10.0.8. Fast Input Reading 12
10.0.9. Bit Hacks 12

1. Data Structures

1.1. Segment Tree.

```
class SegmentTree:
    def __init__(self, data):
        self.n = len(data)
        self.tree = [0] * (2 * self.n)
        for i in range(self.n):
            self.tree[self.n + i] = data[i]
        for i in range(self.n - 1, 0, -1):
            self.tree[i] = self.tree[i << 1] + self.tree[i << 1 | 1]

    def update(self, p, value):
        p += self.n
        self.tree[p] = value
        while p > 1:
            self.tree[p >> 1] = self.tree[p] + self.tree[p ^ 1]
            p >>= 1

    def query(self, l, r):
        res = 0
        l += self.n
        r += self.n
        while l < r:
            if l & 1:
                res += self.tree[l]
                l += 1
            if r & 1:
                r -= 1
                res += self.tree[r]
            l >>= 1
            r >>= 1
        return res
```

1.2. Union Find.

```
class DSU:
    def __init__(self, n):
        self.parent = list(range(n))
        self.size = [1] * n

    def find(self, i):
        if self.parent[i] == i:
            return i
        self.parent[i] = self.find(self.parent[i])
        return self.parent[i]

    def union(self, i, j):
        root_i = self.find(i)
        root_j = self.find(j)

        if root_i != root_j:
            if self.size[root_i] < self.size[root_j]:
                root_i, root_j = root_j, root_i
            self.parent[root_j] = root_i
            self.size[root_i] += self.size[root_j]
            return True
        return False

    def get_size(self, i):
        return self.size[self.find(i)]
```

1.3. 2D Sparse Table.

```
import math

class SparseTable2D:
    def __init__(self, A):
        self.A = A
        self.n = len(A)
        self.m = len(A[0])
```

```
------- self.LOGN = int(math.log2(self.n)) + 1 ------------------------
------- self.LOGM = int(math.log2(self.m)) + 1 ------------------------
 ---------------------------------------------------------------------
------- self.lg = [0] * (max(self.n, self.m) + 1) ---------------------
------- for i in range(2, len(self.lg)): ------------------------------
----------- self.lg[i] = self.lg[i // 2] + 1 -------------------------
 ---------------------------------------------------------------------
------- self.st = [[[[0] * self.m for _ in range(self.n)] ------------
------------------- for _ in range(self.LOGM)] ------------------------
------------------ for _ in range(self.LOGN)] -------------------------
 ---------------------------------------------------------------------
------- self.build() --------------------------------------------------
 ---------------------------------------------------------------------
--- def build(self): ----------------------------------------------------
------- for i in range(self.n): ---------------------------------------
----------- for j in range(self.m): -------------------------------------
--------------- self.st[0][0][i][j] = self.A[i][j] ----------------------
 ---------------------------------------------------------------------
------- for bj in range(1, self.LOGM): --------------------------------
----------- for i in range(self.n): -------------------------------------
--------------- for j in range(self.m - (1 << bj) + 1): -----------------
------------------ self.st[0][bj][i][j] = max( ------------------------
---------------------- self.st[0][bj - 1][i][j], ------------------------
---------------------- self.st[0][bj - 1][i][j + (1 << (bj - 1))] ---
------------------ ) --------------------------------------------------
 ---------------------------------------------------------------------
------- for bi in range(1, self.LOGN): --------------------------------
----------- for i in range(self.n - (1 << bi) + 1): ---------------------
--------------- for j in range(self.m): ---------------------------------
------------------ self.st[bi][0][i][j] = max( ------------------------
---------------------- self.st[bi - 1][0][i][j], ------------------------
---------------------- self.st[bi - 1][0][i + (1 << (bi - 1))][j] ---
------------------ ) --------------------------------------------------
 ---------------------------------------------------------------------
------- for bi in range(1, self.LOGN): --------------------------------
----------- for bj in range(1, self.LOGM): ------------------------------
--------------- for i in range(self.n - (1 << bi) + 1): -----------------
------------------ for j in range(self.m - (1 << bj) + 1): ------------
---------------------- self.st[bi][bj][i][j] = max( -------------------
------------------------- self.st[bi - 1][bj - 1][i][j], --------------
------------------------- self.st[bi - 1][bj - 1][i + (1 << (bi -
↪  1))][j],
------------------------- self.st[bi - 1][bj - 1][i][j + (1 << (bj -
↪  1))],
------------------------- self.st[bi - 1][bj - 1][i + (1 << (bi -
↪  1))][j + (1 << (bj - 1))]
---------------------- ) ----------------------------------------------
 ---------------------------------------------------------------------
--- def query(self, x1, y1, x2, y2): ------------------------------------
------- kx = self.lg[x2 - x1 + 1] -------------------------------------
------- ky = self.lg[y2 - y1 + 1] -------------------------------------
 ---------------------------------------------------------------------
------- return max( ---------------------------------------------------
----------- self.st[kx][ky][x1][y1], ------------------------------------
----------- self.st[kx][ky][x2 - (1 << kx) + 1][y1], --------------------
----------- self.st[kx][ky][x1][y2 - (1 << ky) + 1], --------------------
----------- self.st[kx][ky][x2 - (1 << kx) + 1][y2 - (1 << ky) + 1] --
------- ) -------------------------------------------------------------
# Input Format: st = SparseTable2D(grid) ----------------------------
#               print(st.query(1, 1, 2, 2)) ------------------------
```

2. Dynamic Programming

2.1. Lichao Tree.

```
class Line: ---------------------------------------------------------
--- __slots__ = ['m', 'c'] ------------------------------------------
--- def __init__(self, m=0, c=float('-inf')): -----------------------
------- self.m = m ----------------------------------------------------
------- self.c = c ----------------------------------------------------
------- ---------------------------------------------------------------
--- def eval(self, x): ----------------------------------------------
------- return self.m * x + self.c -------------------------------------
 ---------------------------------------------------------------------
class Node: -----------------------------------------------------------
--- __slots__ = ['line', 'left', 'right'] ---------------------------
--- def __init__(self): ---------------------------------------------
------- self.line = Line() --------------------------------------------
------- self.left = None ----------------------------------------------
------- self.right = None ---------------------------------------------
 ---------------------------------------------------------------------
class LiChaoTree: -----------------------------------------------------
--- def __init__(self, min_x=-10**12, max_x=10**12): ----------------
------- self.root = Node() --------------------------------------------
------- self.min_x = min_x --------------------------------------------
------- self.max_x = max_x --------------------------------------------
 ---------------------------------------------------------------------
--- def add_line(self, m, c): ---------------------------------------
------- self._insert(self.root, self.min_x, self.max_x, Line(m, c)) --
 ---------------------------------------------------------------------
--- def _insert(self, node, l, r, new_line): ------------------------
------- mid = (l + r) // 2 --------------------------------------------
------- ---------------------------------------------------------------
------- left_better = new_line.eval(l) > node.line.eval(l) ------------
------- mid_better = new_line.eval(mid) > node.line.eval(mid) --------
------- ---------------------------------------------------------------
------- if mid_better: ------------------------------------------------
----------- node.line, new_line = new_line, node.line ----------------
------- if r - l <= 1: ------------------------------------------------
----------- return ----------------------------------------------------
----------- ---------------------------------------------------------
------- if left_better != mid_better: ---------------------------------
----------- if not node.left: -----------------------------------------
--------------- node.left = Node() --------------------------------------
----------- self._insert(node.left, l, mid, new_line) ----------------
------- else: ---------------------------------------------------------
----------- if not node.right: ----------------------------------------
--------------- node.right = Node() -------------------------------------
----------- self._insert(node.right, mid, r, new_line) ---------------
 ---------------------------------------------------------------------
--- def query(self, x): ---------------------------------------------
------- return self._query(self.root, self.min_x, self.max_x, x) -----
 ---------------------------------------------------------------------
--- def _query(self, node, l, r, x): --------------------------------
------- if not node: --------------------------------------------------
----------- return float('-inf') --------------------------------------
------- if r - l <= 1: ------------------------------------------------
----------- return node.line.eval(x) ----------------------------------
----------- ---------------------------------------------------------
------- mid = (l + r) // 2 --------------------------------------------
------- res = node.line.eval(x) ---------------------------------------
------- ---------------------------------------------------------------
------- if x < mid: ---------------------------------------------------
----------- res = max(res, self._query(node.left, l, mid, x)) --------
------- else: ---------------------------------------------------------
----------- res = max(res, self._query(node.right, mid, r, x)) -------
----------- ---------------------------------------------------------
------- return res ----------------------------------------------------
```

2.2. Divide And Conquer.

```
INF = float('inf') --------------------------------------------------
 ---------------------------------------------------------------------
def solve_dp(g, k_L, k_R, n_L, n_R, dp, cost_func): -------------------
--- if n_L > n_R: ---------------------------------------------------
------- return --------------------------------------------------------
 ---------------------------------------------------------------------
--- n_M = (n_L + n_R) // 2 ------------------------------------------
--- best_k = -1 -----------------------------------------------------
--- dp[g][n_M] = INF ------------------------------------------------
 ---------------------------------------------------------------------
--- for k in range(k_L, min(n_M, k_R) + 1): -------------------------
------- current_val = dp[g - 1][k] + cost_func(k + 1, n_M) ----------
------- if current_val < dp[g][n_M]: ----------------------------------
----------- dp[g][n_M] = current_val ----------------------------------
----------- best_k = k ------------------------------------------------
 ---------------------------------------------------------------------
--- solve_dp(g, k_L, best_k, n_L, n_M - 1, dp, cost_func) ------------
--- solve_dp(g, best_k, k_R, n_M + 1, n_R, dp, cost_func) ------------
```

3. Geometry

3.1. Angles And Rotations.

```
def angle(a, b, c): # angle formed by abc in radians: PI < x <= PI ---
--- return abs((arg(a-b) - arg(c-b)) % (2 * math.pi)) ----------------
def rotate(p, a, d): # rotate point a about pivot p CCW at d radians -
--- return p + (a - p) * complex(math.cos(d), math.sin(d)) -----------
```

3.2. Distances.

```
 ---------------------------------------------------------------------
# Distance: Point to 2D Line (ax + by + c = 0) ----------------------
def dist_pt_line_eq(p, a, b, c): ------------------------------------
--- return abs(a*p.real + b*p.imag + c) / math.hypot(a, b) -----------
 ---------------------------------------------------------------------
# Distance: Point to 2D Line through points a and b ------------------
def dist_pt_line(p, a, b): ------------------------------------------
--- cp = (p - a).conjugate() * (b - a) ------------------------------
--- return abs(cp.imag) / abs(b - a) --------------------------------
 ---------------------------------------------------------------------
# Distance: Point to 3D Plane (ax + by + cz + d = 0) ----------------
def dist_pt_plane(p, a, b, c, d): -----------------------------------
--- # math.hypot supports 3+ arguments in Python 3.8+ ----------------
--- return abs(a*p.x + b*p.y + c*p.z + d) / math.hypot(a,b,c) --------
 ---------------------------------------------------------------------
# Distance: Shortest distance between 3D lines AB and CD ------------
def dist_line_3d(A, B, C, D): ---------------------------------------
--- u, v, w = B - A, D - C, A - C -----------------------------------
--- a, b = u.dot(u), u.dot(v) ---------------------------------------
--- c, d, e = v.dot(v), u.dot(w), v.dot(w) --------------------------
--- det = a * c - b * b ---------------------------------------------
--- if det < 1e-9: # Parallel lines ---------------------------------
------- s, t = 0.0, d / b if b > c else e / c -----------------------
--- else: # Skew lines ----------------------------------------------
------- s = (b * e - c * d) / det -----------------------------------
------- t = (a * e - b * d) / det -----------------------------------
--- p1 = P3(A.x + u.x*s, A.y + u.y*s, A.z + u.z*s) ------------------
--- p2 = P3(C.x + v.x*t, C.y + v.y*t, C.z + v.z*t) ------------------
--- vec = p1 - p2 ---------------------------------------------------
--- return math.sqrt(vec.dot(vec)) ----------------------------------
```

3.3. Intersections.

3.3.1. *Circle Line Intersection.*

```
EPS = 1e-9 ----------------------------------------------------------
def projLine(p, a, b): # Project point p onto line segment ab --------
--- v = b - a -------------------------------------------------------
--- scalar = ((p - a) * v.conjugate()).real / abs(v)**2 -------------
--- return a + v * scalar -------------------------------------------
 ---------------------------------------------------------------------
def rotate(c, p, t): # Rotate point p around center c by t radians ---
--- return c + (p - c) * cmath.rect(1, t) ---------------------------
 ---------------------------------------------------------------------
def CL_inter(c, r, a, b): -------------------------------------------
--- p = projLine(c, a, b) -------------------------------------------
--- d = abs(c - p) --------------------------------------------------
--- ans = [] --------------------------------------------------------
--- ---------------------------------------------------------------
--- if d > r + EPS: pass # None -------------------------------------
--- elif d > r - EPS: ans.append(p) # Tangent ----------------------
--- elif d < EPS: # Diameter ----------------------------------------
------- v = r * (b - a) / abs(b - a) --------------------------------
------- ans.append(c + v) -------------------------------------------
------- ans.append(c - v) -------------------------------------------
```

```
--- else: # Secant ----------
------- t = math.acos(d / r) ----------
------- p = c + (p - c) * r / d ----------
------- ans.append(rotate(c, p, t)) ----------
------- ans.append(rotate(c, p, -t)) ----------
--- return ans ----------
```

3.3.2. *Circle Circle Intersection.*

```
EPS = 1e-9 ----------
def rotate(c, p, t): ----------
--- # Rotate point p around center c by t radians ----------
--- return c + (p - c) * cmath.rect(1, t) ----------
 ----------
def CC_inter(c1, r1, c2, r2): ----------
--- d = abs(c1 - c2) ----------
--- ans = [] ----------
--- if d < EPS: # Concentric circles ----------
------- if abs(r1 - r2) < EPS: ----------
----------- pass # Infinite intersections (same circle) ----------
--- elif r1 < EPS: # Circle 1 is a point ----------
------- if abs(d - r2) < EPS: ----------
----------- ans.append(c1) ----------
--- else: # General case using Law of Cosines ----------
------- s = (r1*r1 + d*d - r2*r2) / (2*r1*d) ----------
------- s = max(-1.0, min(1.0, s)) # Prevent float domain errors -----
------- t = math.acos(s) ----------
------- # Point on C1's radius pointing towards C2 ----------
------- mid = c1 + (c2 - c1) * r1 / d ----------
------- ans.append(rotate(c1, mid, t)) ----------
------- if abs(math.sin(t)) >= EPS: ----------
----------- # FIXED: Rotate around c1, not c2! ----------
----------- ans.append(rotate(c1, mid, -t)) ----------
--- return ans ----------
```

3.3.3. *Line Intersect.*

```
def line_inter_fast(a, b, c, d): ----------
--- A = cross(d - a, b - a) ----------
--- B = cross(c - a, b - a) ----------
--- return (B * d - A * c) / (B - A) if B - A else None ----------
```

3.4. Closest Pair.

```
def closest_pair(pts): ----------
--- if len(pts) <= 1: return float('inf') ----------
--- px = sorted(pts) # Automatically sorts by x, then y ----------
--- def solve(l, r): # Base case: Brute force for small partitions ---
------- if r - l <= 3: ----------
----------- d = float('inf') ----------
----------- for i in range(l, r): ----------
--------------- for j in range(i + 1, r): ----------
------------------- d = min(d, math.dist(px[i], px[j])) ----------
----------- return d ----------
------- mid = (l + r) // 2 ----------
------- mid_x = px[mid][0] ----------
------- # Recursively find the minimum in both halves ----------
------- d = min(solve(l, mid), solve(mid, r)) ----------
------- # Gather points in the middle strip ----------
------- strip = [p for p in px[l:r] if abs(p[0] - mid_x) < d] --------
------- strip.sort(key=lambda p: p[1]) # Sort strip by y ----------
------- # Check at most 7 geometric neighbors in the strip ----------
------- for i, p1 in enumerate(strip): ----------
----------- for p2 in strip[i+1 : i+8]: ----------
--------------- d = min(d, math.dist(p1, p2)) ----------
------- return d ----------
--- return solve(0, len(px)) ----------
```

3.5. Convex Hull.

3.5.1. *Convex Hull.*

```
def cross(o, a, b): # 2D cross product of OA and OB vectors ----------
--- return (a[0]-o[0])*(b[1]-o[1]) - (a[1]-o[1])*(b[0]-o[0]) ---------
 ----------
def convex_hull(pts): # Sorts by x, then by y automatically ----------
--- pts = sorted(set(pts)) ----------
--- if len(pts) <= 1: ----------
------- return pts ----------
--- def build(iterable): ----------
------- h = [] ----------
------- for p in iterable: ----------
----------- while len(h) >= 2 and cross(h[-2], h[-1], p) <= 0: -------
--------------- h.pop() ----------
----------- h.append(p) ----------
------- return h ----------
--- lower = build(pts) ----------
--- upper = build(reversed(pts)) ----------
--- return lower[:-1] + upper[:-1] ----------
```

3.5.2. *Convex Hull 3D.*

```
class P3: ----------
--- __slots__ = ['x', 'y', 'z'] ----------
--- def __init__(self, x=0, y=0, z=0): ----------
------- self.x, self.y, self.z = x, y, z ----------
--- def __sub__(self, o): ----------
------- return P3(self.x-o.x, self.y-o.y, self.z-o.z) ----------
--- def cross(self, o): ----------
------- return P3(self.y*o.z - self.z*o.y, ----------
---------------- self.z*o.x - self.x*o.z, ----------
---------------- self.x*o.y - self.y*o.x) ----------
--- def dot(self, o): ----------
------- return self.x*o.x + self.y*o.y + self.z*o.z ----------
--- def __eq__(self, o): ----------
------- return (self.x, self.y, self.z) == (o.x, o.y, o.z) ----------
--- def __lt__(self, o): ----------
------- return (self.x, self.y, self.z) < (o.x, o.y, o.z) ----------
 ----------
class Face: ----------
--- __slots__ = ['p', 'q'] ----------
--- def __init__(self, p, q): ----------
------- self.p = p # Tuple of 3 point indices ----------
------- self.q = q # Normal vector (P3 object) ----------
 ----------
def hull_3D(pts): ----------
--- n = len(pts) ----------
--- faces = [] ----------
--- dead = [[True] * n for _ in range(n)] ----------
--- def add_face(a, b, c): ----------
------- q = (pts[b] - pts[a]).cross(pts[c] - pts[a]) ----------
------- faces.append(Face((a, b, c), q)) ----------
------- dead[a][b] = dead[b][c] = dead[c][a] = False ----------
--- # Initialize with a degenerate tetrahedron ----------
--- add_face(0, 1, 2) ----------
--- add_face(0, 2, 1) ----------
--- for i in range(3, n): ----------
------- f_inv = [] ----------
------- for f in faces: ----------
----------- # If face is visible from the point ----------
----------- if (pts[i] - pts[f.p[0]]).dot(f.q) > 0: ----------
--------------- for j in range(3): ----------
------------------- dead[f.p[j]][f.p[(j+1)%3]] = True ----------
----------- else: ----------
--------------- f_inv.append(f) ----------
------- faces = [] ----------
------- for f in f_inv: ----------
----------- for j in range(3): ----------
--------------- a, b = f.p[j], f.p[(j+1)%3] ----------
--------------- # If reverse edge is dead but forward is alive -------
--------------- if dead[b][a]: ----------
------------------- add_face(b, a, i) ----------
------- faces.extend(f_inv) ----------
--- return faces ----------
```

3.6. Areas.

3.6.1. *Triangle Area.*

```
def area(a, b, c): ----------
--- s = (a + b + c) / 2 ----------
--- return (s*(s-a)*(s-b)*(s-c)) ** 0.5 ----------
```

3.6.2. *Polygon Area.*

```
def area(p, n): ----------
--- a = 0 ----------
--- for i in range(n): ----------
------- j = (i - 1) % n ----------
------- a += cross(p[i], p[j]) ----------
--- return abs(a) / 2 ----------
```

3.7. Dot Cross.

```
def dot(a, b): ----------
--- return a.x * b.x + a.y * b.y # + a.z * b.z ----------
def cross(a, b): ----------
--- return a.x * b.y - a.y * b.x ----------
def cross(a, b, c): ----------
--- return cross(a, b) + cross(b, c) + cross(c, a) ----------
def cross3D(a, b): ----------
--- return point3(a.x*b.y - a.y*b.x, a.y*b.z - a.z*b.y, a.z*b.x -
↪  a.x*b.z)
```

3.8. Line Sweep Closest Pair.

```
import math ----------
 ----------
def dist_sq(p1, p2): ----------
--- return (p1[0] - p2[0])**2 + (p1[1] - p2[1])**2 ----------
 ----------
 ----------
def closest_pair(points): ----------
--- points.sort() ----------
 ----------
--- def solve(pts): ----------
------- n = len(pts) ----------
 ----------
------- if n <= 3: ----------
----------- return min(dist_sq(pts[i], pts[j]) ----------
--------------------- for i in range(n) ----------
--------------------- for j in range(i + 1, n)) if n > 1 else
↪  float('inf')
 ----------
------- mid = n // 2 ----------
------- mid_x = pts[mid][0] ----------
 ----------
------- left = pts[:mid] ----------
------- right = pts[mid:] ----------
 ----------
------- d = min(solve(left), solve(right)) ----------
 ----------
------- strip = [p for p in pts if (p[0] - mid_x) ** 2 < d] ----------
------- strip.sort(key=lambda p: p[1]) ----------
------- ----------
------- for i in range(len(strip)): ----------
----------- for j in range(i + 1, min(i + 7, len(strip))): ----------
--------------- d = min(d, dist_sq(strip[i], strip[j])) ----------
 ----------
------- return d ----------
 ----------
--- return solve(points) ----------
```

4. Graphs

4.1. Max Flow.

4.1.1. *Dinic.*

```
INF = float('inf')
class Dinic:
    def __init__(self, n, s, t):
        self.n, self.s, self.t = n, s, t
        self.edges = []
        self.adj = [[] for _ in range(n)]

    def add_edge(self, u, v, c, bi=False):
        self.adj[u].append(len(self.edges))
        self.edges.append([v, c, 0])
        self.adj[v].append(len(self.edges))
        self.edges.append([u, c if bi else 0, 0])

    def bfs(self):
        self.dist = [-1] * self.n
        self.dist[self.s] = 0
        q = deque([self.s])
        while q:
            u = q.popleft()
            for i in self.adj[u]:
                v, c, f = self.edges[i]
                if self.dist[v] < 0 and c > f:
                    self.dist[v] = self.dist[u] + 1
                    q.append(v)
        return self.dist[self.t] != -1
    
    def dfs(self, u, pushed):
        if pushed == 0 or u == self.t: return pushed
        while self.ptr[u] < len(self.adj[u]):
            i = self.adj[u][self.ptr[u]]
            v, c, f = self.edges[i]
            if self.dist[u]+1 == self.dist[v] and c > f:
                tr = self.dfs(v, min(pushed, c - f))
                if tr > 0:
                    self.edges[i][2] += tr
                    self.edges[i^1][2] -= tr
                    return tr
            self.ptr[u] += 1
        return 0

    def calc_max_flow(self):
        flow = 0
        while self.bfs():
            self.ptr = [0] * self.n
            while True:
                pushed = self.dfs(self.s, INF)
                if not pushed:
                    break
                flow += pushed
        return flow
```

4.1.2. *Edmonds Karp.*

```
INF = float('inf')
class EdmondsKarp:
    def __init__(self, n, s, t):
        self.n, self.s, self.t = n, s, t
        self.adj = [[] for _ in range(n)]
        self.res = [[0] * n for _ in range(n)]

    def add_edge(self, u, v, w):
        self.adj[u].append(v)
        self.adj[v].append(u)
        self.res[u][v] += w

    def bfs(self):
        self.par = [-1] * self.n
        self.par[self.s] = self.s
        q = deque([self.s])
        while q:
            u = q.popleft()
            for v in self.adj[u]:
                if self.res[u][v] > 0 and self.par[v] == -1:
                    self.par[v] = u
                    if v == self.t:
                        return True
                    q.append(v)
        return False

    def calc_max_flow(self):
        ans = 0
        while self.bfs():
            flow = INF
            u = self.t
            while u != self.s:
                p = self.par[u]
                flow = min(flow, self.res[p][u])
                u = p
            u = self.t
            while u != self.s:
                p = self.par[u]
                self.res[p][u] -= flow
                self.res[u][p] += flow
                u = p
            ans += flow
        return ans
```

4.1.3. *Bipartite Matching.*

```
from collections import deque

INF = float('inf')
class BipartiteGraph:
    def __init__(self, n, m):
        self.n = n
        self.m = m
        self.adj = [[] for _ in range(n)]
        self.L = [-1] * n
        self.R = [-1] * m
        self.dist = [INF] * (n + 1)

    def add_edge(self, u, v):
        self.adj[u].append(v)

    def bfs(self):
        queue = deque()
        for v in range(self.n):
            if self.L[v] == -1:
                self.dist[v] = 0
                queue.append(v)
            else:
                self.dist[v] = INF

        self.dist[self.n] = INF

        while queue:
            v = queue.popleft()
            if self.dist[v] < self.dist[self.n]:
                for u in self.adj[v]:
                    next_v = self.R[u]
                    dist_next = self.dist[next_v] if next_v != -1 else
↪   self.dist[self.n]

                    if (next_v == -1 and self.dist[self.n] == INF) or \
                        (next_v != -1 and self.dist[next_v] ==
↪   INF):

                        target = next_v if next_v != -1 else self.n --
                        self.dist[target] = self.dist[v] + 1 --------
                        if target != self.n: ------------------------
                            queue.append(target) ------------------

        return self.dist[self.n] != INF

    def dfs(self, v):
        if v != -1:
            for u in self.adj[v]:
                next_v = self.R[u]
                dist_next = self.dist[next_v] if next_v != -1 else
↪   self.dist[self.n]

                if dist_next == self.dist[v] + 1:
                    if self.dfs(next_v):
                        self.R[u] = v
                        self.L[v] = u
                        return True
            self.dist[v] = INF
            return False
        return True

    def maximum_matching(self):
        matching = 0
        for u in range(self.n):
            for v in self.adj[u]:
                if self.R[v] == -1:
                    self.L[u] = v
                    self.R[v] = u
                    matching += 1
                    break

        while self.bfs():
            for i in range(self.n):
                if self.L[i] == -1 and self.dfs(i):
                    matching += 1
        return matching
# Input Format: g = BipartiteGraph(2, 2)
#               g.add_edge(0, 0)
#               g.add_edge(0, 1)
#               g.add_edge(1, 0)
#               print(f"Max Matching: {g.maximum_matching()}")
```

4.2. Shortest Path.

4.2.1. *Dijkstra.*

```
import heapq
INF = float('inf')
def dijkstra(n, adj, s):
    dist = [INF] * n
    dist[s] = 0
    pq = [(0, s)]

    while pq:
        d, u = heapq.heappop(pq)
        if dist[u] < d:
            continue
        for v, w in adj[u]:
            if dist[v] > d + w:
                dist[v] = d + w
                heapq.heappush(pq, (dist[v], v))
    return dist
```

4.2.2. *Bellman Ford.*

```
INF = float('inf')
def bellman_ford(n, adj, s):
    dist = [INF] * n
    dist[s] = 0
    for _ in range(n - 1):
        for u in range(n):
            if dist[u] == INF: continue
            for v, w in adj[u]:
```

```
---------------- if dist[u] + w < dist[v]:
-------------------- dist[v] = dist[u] + w
--- for u in range(n):
------- if dist[u] == INF: continue
------- for v, w in adj[u]:
----------- if dist[u] + w < dist[v]:
---------------- return dist, True
--- return dist, False
```

4.3. Minimum Spanning Tree.

4.3.1. *Kruskal Min Spanning Tree.*

```
import heapq

class UnionFind:
--- def __init__(self, n):
------- self.parent = list(range(n))
------- self.rank = [0] * n

--- def find(self, x):
------- while self.parent[x] != x:
----------- self.parent[x] = self.parent[self.parent[x]]
----------- x = self.parent[x]
------- return x

--- def unite(self, a, b):
------- a = self.find(a)
------- b = self.find(b)

------- if a == b:
----------- return False

------- if self.rank[a] < self.rank[b]:
----------- self.parent[a] = b
------- else:
----------- self.parent[b] = a
----------- if self.rank[a] == self.rank[b]:
---------------- self.rank[a] += 1

------- return True

def kruskal(n, edges):
--- edges = edges[:]
--- heapq.heapify(edges)

--- uf = UnionFind(n)
--- total_weight = 0
--- mst = []

--- while edges:
------- w, u, v = heapq.heappop(edges)
------- if uf.unite(u, v):
----------- mst.append((u, v, w))
----------- total_weight += w

----------- if len(mst) == n - 1:
---------------- break

--- return mst, total_weight
```

4.3.2. *Prim Min Spanning Tree.*

```
import heapq

def prim(n, adj, start=0):
--- visited = [False] * n
--- pq = [(0, start, -1)]

--- total_weight = 0
--- mst = []

--- while pq:
------- w, u, parent = heapq.heappop(pq)

------- if visited[u]:
----------- continue

------- visited[u] = True

------- if parent != -1:
----------- mst.append((parent, u, w))
----------- total_weight += w

------- for v, weight in adj[u]:
----------- if not visited[v]:
---------------- heapq.heappush(pq, (weight, v, u))

--- return mst, total_weight
```

4.4. Least Common Ancestor.

4.4.1. *Binary Lifting Lca.*

```
import math

class BinaryLifting:
--- def __init__(self, adj):
------- self.n = len(adj)
------- self.LOG = math.ceil(math.log2(self.n)) + 1
------- self.adj = adj

------- self.dep = [0] * self.n
------- self.par = [[-1] * self.LOG for _ in range(self.n)]

------- self.build(0)

--- def dfs(self, u, p):
------- self.par[u][0] = p
------- for v in self.adj[u]:
----------- if v != p:
---------------- self.dep[v] = self.dep[u] + 1
---------------- self.dfs(v, u)

--- def build(self, root):
------- self.dfs(root, root)

------- for k in range(1, self.LOG):
----------- for u in range(self.n):
---------------- self.par[u][k] = self.par[self.par[u][k-1]][k-1]

--- def lift(self, u, k):
------- for i in range(self.LOG):
----------- if k & (1 << i):
---------------- u = self.par[u][i]
------- return u

--- def lca(self, u, v):
------- if self.dep[u] < self.dep[v]:
----------- u, v = v, u

------- u = self.lift(u, self.dep[u] - self.dep[v])

------- if u == v:
----------- return u

------- for k in reversed(range(self.LOG)):
----------- if self.par[u][k] != self.par[v][k]:
---------------- u = self.par[u][k]
---------------- v = self.par[v][k]

------- return self.par[u][0]
```

4.4.2. *Euler Lca.*

```
class EulerLCA:
--- def __init__(self, adj):
------- self.n = len(adj)
------- self.adj = adj

------- self.euler = []
------- self.first = [-1] * self.n
------- self.depth = [0] * self.n

------- self.build(0)

--- def dfs(self, u, p, d):
------- self.first[u] = len(self.euler)
------- self.depth[u] = d
------- self.euler.append(u)

------- for v in self.adj[u]:
----------- if v != p:
---------------- self.dfs(v, u, d + 1)
---------------- self.euler.append(u)

--- def build(self, root):
------- self.dfs(root, root, 0)

------- m = len(self.euler)
------- LOG = m.bit_length()

------- self.st = [[0] * m for _ in range(LOG)]
------- self.log = [0] * (m + 1)

------- for i in range(2, m + 1):
----------- self.log[i] = self.log[i // 2] + 1

------- for i in range(m):
----------- self.st[0][i] = self.euler[i]

------- for k in range(1, LOG):
----------- for i in range(m - (1 << k) + 1):
---------------- a = self.st[k-1][i]
---------------- b = self.st[k-1][i + (1 << (k-1))]
---------------- self.st[k][i] = a if self.depth[a] < self.depth[b] else
↪  b

--- def lca(self, u, v):
------- l, r = self.first[u], self.first[v]
------- if l > r:
----------- l, r = r, l

------- k = self.log[r - l + 1]
------- a = self.st[k][l]
------- b = self.st[k][r - (1 << k) + 1]

------- return a if self.depth[a] < self.depth[b] else b
```

4.4.3. *Tarjan Lca.*

```
class DSU:
--- def __init__(self, n):
------- self.p = list(range(n))

--- def find(self, x):
------- if self.p[x] != x:
----------- self.p[x] = self.find(self.p[x])
------- return self.p[x]

--- def unite(self, a, b):
------- self.p[self.find(b)] = self.find(a)

class Tarjan:
--- def __init__(self, adj, queries):
```

```
        self.adj = adj
        self.queries = queries
        self.n = len(adj)

        self.dsu = DSU(self.n)
        self.vis = [False] * self.n
        self.anc = [0] * self.n
        self.res = {}

    def dfs(self, u, p):
        self.anc[u] = u

        for v in self.adj[u]:
            if v == p:
                continue
            self.dfs(v, u)
            self.dsu.unite(u, v)
            self.anc[self.dsu.find(u)] = u

        self.vis[u] = True

        for v, idx in self.queries[u]:
            if self.vis[v]:
                self.res[idx] = self.anc[self.dsu.find(v)]
```

4.5. Minimum Mean Weight Cycle.

```
def min_mean_cycle(adj):
    n = len(adj)
    INF = float('inf')

    arr = [[INF] * n for _ in range(n + 1)]

    for i in range(n):
        arr[0][i] = 0

    for k in range(1, n + 1):
        for u in range(n):
            if arr[k - 1][u] == INF:
                continue
            for v, w in adj[u]:
                arr[k][v] = min(arr[k][v], arr[k - 1][u] + w)

    result = INF

    for v in range(n):
        if arr[n][v] == INF:
            continue

        max_avg = -INF
        for k in range(n):
            if arr[k][v] == INF:
                continue
            avg = (arr[n][v] - arr[k][v]) / (n - k)
            max_avg = max(max_avg, avg)

        result = min(result, max_avg)

    return result
```

4.6. Strongly Connected Components.

```
class KosarajuGraph:
    def __init__(self, n):
        self.n = n
        self.graph = [[] for _ in range(n)]
        self.rev_graph = [[] for _ in range(n)]
        self.vis = [0] * n
        self.sccs = []

    def add_edge(self, u, v):
        self.graph[u].append(v)
        self.rev_graph[v].append(u)

    def dfs(self, u, adj, container):
        self.vis[u] = 1
        for v in adj[u]:
            if not self.vis[v]:
                self.dfs(v, adj, container)
        container.append(u)

    def kosaraju(self):
        topo = []

        # First pass
        self.vis = [0] * self.n
        for i in range(self.n):
            if not self.vis[i]:
                self.dfs(i, self.graph, topo)

        # Second pass
        self.vis = [0] * self.n
        self.sccs = []

        for u in reversed(topo):
            if not self.vis[u]:
                comp = []
                self.dfs(u, self.rev_graph, comp)
                self.sccs.append(comp)

        return self.sccs
```

4.7. Heavy Light Decomposition.

```
class SegmentTree:
    def __init__(self, n):
        self.n = n
        self.tree = [0] * (4 * n)

    # Example: point update
    def update(self, idx, val, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if l == r:
            self.tree[node] = vala
            return
        mid = (l + r) // 2
        if idx <= mid:
            self.update(idx, val, node * 2, l, mid)
        else:
            self.update(idx, val, node * 2 + 1, mid + 1, r)
        self.tree[node] = self.tree[node * 2] + self.tree[node * 2 + 1]

    # Example: range query
    def query(self, ql, qr, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if qr < l or r < ql:
            return 0
        if ql <= l and r <= qr:
            return self.tree[node]
        mid = (l + r) // 2
        return self.query(ql, qr, node * 2, l, mid) + \
               self.query(ql, qr, node * 2 + 1, mid + 1, r)

class HeavyLightTree:
    def __init__(self, n):
        self.n = n
        self.adj = [[] for _ in range(n)]

        self.par = [-1] * n
        self.heavy = [-1] * n
        self.dep = [0] * n
        self.path_root = [0] * n
        self.pos = [0] * n

        self.segment_tree = SegmentTree(n)
        self.current_pos = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.adj[v].append(u)

    def dfs(self, u):
        size = 1
        max_subtree_size = 0

        for v in self.adj[u]:
            if v != self.par[u]:
                self.par[v] = u
                self.dep[v] = self.dep[u] + 1
                subtree_size = self.dfs(v)

                if subtree_size > max_subtree_size:
                    max_subtree_size = subtree_size
                    self.heavy[u] = v

                size += subtree_size

        return size

    def decompose(self, u, root):
        self.path_root[u] = root
        self.pos[u] = self.current_pos
        self.current_pos += 1

        if self.heavy[u] != -1:
            self.decompose(self.heavy[u], root)

        for v in self.adj[u]:
            if v != self.par[u] and v != self.heavy[u]:
                self.decompose(v, v)

    def build(self, root=0):
        self.par[root] = root
        self.dep[root] = 0
        self.dfs(root)
        self.decompose(root, root)

    # Query path u -> v
    def query(self, u, v):
        res = 0
        while self.path_root[u] != self.path_root[v]:
            if self.dep[self.path_root[u]] <
↪   self.dep[self.path_root[v]]:
                u, v = v, u
            start = self.path_root[u]
            res += self.segment_tree.query(self.pos[start],
↪   self.pos[u])
            u = self.par[start]

        # Same path
        if self.dep[u] > self.dep[v]:
            u, v = v, u
        res += self.segment_tree.query(self.pos[u], self.pos[v])
        return res

    # Update node value
    def update(self, u, value):
        self.segment_tree.update(self.pos[u], value)
```

4.8. Euler Path/Cycle.

4.8.1. *Directed Euler.*

```
from collections import defaultdict

class DirectedEuler:
    def __init__(self, n):
        self.n = n
        self.adj = defaultdict(list)
        self.indeg = [0]*n
        self.outdeg = [0]*n
        self.edge_count = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.outdeg[u] += 1
        self.indeg[v] += 1
        self.edge_count += 1

    def find_start_node(self):
        start = -1
        end = -1
        for i in range(self.n):
            if self.outdeg[i] - self.indeg[i] == 1:
                if start != -1:
                    return -1
                start = i
            elif self.indeg[i] - self.outdeg[i] == 1:
                if end != -1:
                    return -1
                end = i
            elif self.indeg[i] != self.outdeg[i]:
                return -1

        if start == -1:
            for i in range(self.n):
                if self.outdeg[i] > 0:
                    return i
        return start

    def get_euler_path(self):
        start = self.find_start_node()
        if start == -1:
            return None

        stack = [start]
        path = []

        local_adj = {u: self.adj[u][:] for u in self.adj}

        while stack:
            u = stack[-1]
            if u in local_adj and local_adj[u]:
                v = local_adj[u].pop()
                stack.append(v)
            else:
                path.append(stack.pop())

        if len(path) != self.edge_count + 1:
            return None

        return path[::-1]


    class UndirectedEuler:
        def __init__(self, n):
            self.n = n
            self.adj = defaultdict(list)
            self.edge_count = 0

        def add_edge(self, u, v):
            self.adj[u].append(v)
            self.adj[v].append(u)
            self.edge_count += 1

        def is_valid(self):
            odd = 0
            for i in range(self.n):
                if len(self.adj[i]) % 2 != 0:
                    odd += 1
            return odd == 0 or odd == 2

        def get_euler_path(self):
            if not self.is_valid():
                return None

            start = 0
            for i in range(self.n):
                if len(self.adj[i]) % 2 == 1:
                    start = i
                    break
                if len(self.adj[i]) > 0:
                    start = i

            stack = [start]
            path = []

            local_adj = {u: self.adj[u][:] for u in self.adj}

            while stack:
                u = stack[-1]
                if u in local_adj and local_adj[u]:
                    v = local_adj[u].pop()
                    local_adj[v].remove(u)
                    stack.append(v)
                else:
                    path.append(stack.pop())

            if len(path) != self.edge_count + 1:
                return None

            return path[::-1]
```

4.8.2. *Undirected Euler.*

```
from collections import defaultdict

class UndirectedEuler:
    def __init__(self, n):
        self.n = n
        self.adj = defaultdict(list)
        self.edge_count = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.adj[v].append(u)
        self.edge_count += 1

    def is_valid(self):
        odd = 0
        for i in range(self.n):
            if len(self.adj[i]) % 2 != 0:
                odd += 1
        return odd == 0 or odd == 2

    def get_euler_path(self):
        if not self.is_valid():
            return None

        start = 0
        for i in range(self.n):
            if len(self.adj[i]) % 2 == 1:
                start = i
                break
            if len(self.adj[i]) > 0:
                start = i

        stack = [start]
        path = []

        local_adj = {u: self.adj[u][:] for u in self.adj}

        while stack:
            u = stack[-1]
            if u in local_adj and local_adj[u]:
                v = local_adj[u].pop()
                local_adj[v].remove(u)
                stack.append(v)
            else:
                path.append(stack.pop())

        if len(path) != self.edge_count + 1:
            return None

        return path[::-1]
```

4.9. Tree Isomorphism.

```
from collections import deque

def bfs(u, adj, n):
    vis = [False] * n
    pre = [-1] * n
    q = deque()

    q.append(u)
    vis[u] = True

    last = u

    while q:
        u = q.popleft()
        last = u

        for v in adj[u]:
            if not vis[v]:
                vis[v] = True
                pre[v] = u
                q.append(v)

    return last, pre

def tree_centers(r, adj, n):
    u, _ = bfs(r, adj, n)
    v, pre = bfs(u, adj, n)

    path = []
    while v != -1:
        path.append(v)
        v = pre[v]

    size = len(path)

    med = [path[size // 2]]
    if size % 2 == 0:
        med.append(path[size // 2 - 1])

    return med

def rootcode(u, adj, p, d, pr):
    children = []
    for v in adj[u]:
        if v != p:
            children.append(rootcode(v, adj, u, (d + 1) % len(pr), pr))

    children.sort()

    h = len(children) + 1
    for x in children:
```

```
------- h = h * pr[d] + x ----------------------------------------------
--------------------------------------------------------------------------
--- return h -------------------------------------------------------------
--------------------------------------------------------------------------
def treecode(root, adj, n, pr): ----------------------------------------
--- c = tree_centers(root, adj, n) -----------------------------------------
--------------------------------------------------------------------------
--- if len(c) == 1: --------------------------------------------------------
------- return (rootcode(c[0], adj, -1, 0, pr) << 1) | 1 -----------------
--- else: ------------------------------------------------------------------
------- return (rootcode(c[0], adj, -1, 0, pr) * ---------------------------
--------------- rootcode(c[1], adj, -1, 0, pr)) << 1 ----------------------
--------------------------------------------------------------------------
def isomorphic(r1, adj1, r2, adj2, n1, n2, pr, rooted=False): ----------
--- if rooted: -------------------------------------------------------------
------- return rootcode(r1, adj1, -1, 0, pr) == rootcode(r2, adj2, -1,
↪  0, pr)
--- return treecode(r1, adj1, n1, pr) == treecode(r2, adj2, n2, pr) --
```

4.10. Centroid Decomposition.

```
import sys -------------------------------------------------------------
sys.setrecursionlimit(10**7) -------------------------------------------
--------------------------------------------------------------------------
INF = 10**9 ------------------------------------------------------------
--------------------------------------------------------------------------
class CentroidDecomposition: -------------------------------------------
--- def __init__(self, n): -------------------------------------------
------- self.n = n -------------------------------------------------
------- self.adj = [[] for _ in range(n)] --------------------------
--------------------------------------------------------------------
------- self.LG = 20 -----------------------------------------------
------- self.jmp = [[-1] * self.LG for _ in range(n)] --------------
------- self.path = [[INF] * self.LG for _ in range(n)] ------------
--------------------------------------------------------------------
------- self.sz = [0] * n ------------------------------------------
------- self.seph = [0] * n ----------------------------------------
------- self.shortest = [INF] * n ----------------------------------
------- self.blocked = [False] * n ---------------------------------
--------------------------------------------------------------------
--- def add_edge(self, a, b): ----------------------------------------
------- self.adj[a].append(b) --------------------------------------
------- self.adj[b].append(a) --------------------------------------
--------------------------------------------------------------------
--- def dfs(self, u, p): ---------------------------------------------
------- self.sz[u] = 1 ---------------------------------------------
------- for v in self.adj[u]: --------------------------------------
----------- if v != p and not self.blocked[v]: -------------------
--------------- self.sz[u] += self.dfs(v, u) -------------------
------- return self.sz[u] ------------------------------------------
--------------------------------------------------------------------
--- def find_centroid(self, u, p, n): --------------------------------
------- for v in self.adj[u]: --------------------------------------
----------- if v != p and not self.blocked[v] and self.sz[v] > n // 2:
--------------- return self.find_centroid(v, u, n) -------------------
------- return u ---------------------------------------------------
--------------------------------------------------------------------
--- def makepaths(self, sep, u, p, depth, h): ------------------------
------- self.jmp[u][h] = sep ---------------------------------------
------- self.path[u][h] = depth ------------------------------------
--------------------------------------------------------------------
------- for v in self.adj[u]: --------------------------------------
----------- if v != p and not self.blocked[v]: -------------------
--------------- self.makepaths(sep, v, u, depth + 1, h) --------------
--------------------------------------------------------------------
--- def separate(self, h=0, u=0): ------------------------------------
------- self.dfs(u, -1) --------------------------------------------
------- c = self.find_centroid(u, -1, self.sz[u]) ------------------
--------------------------------------------------------------------
------- self.seph[c] = h -------------------------------------------
------- self.makepaths(c, c, -1, 0, h) -----------------------------
--------------------------------------------------------------------
------- self.blocked[c] = True -------------------------------------
--------------------------------------------------------------------
------- for v in self.adj[c]: --------------------------------------
----------- if not self.blocked[v]: ------------------------------
--------------- self.separate(h + 1, v) ------------------------------
--------------------------------------------------------------------
--- def paint(self, u): ----------------------------------------------
------- for h in range(self.seph[u] + 1): --------------------------
----------- c = self.jmp[u][h] -----------------------------------
----------- self.shortest[c] = min(self.shortest[c], self.path[u][h])
--------------------------------------------------------------------
--- def closest(self, u): --------------------------------------------
------- ans = INF --------------------------------------------------
------- for h in range(self.seph[u] + 1): --------------------------
----------- c = self.jmp[u][h] -----------------------------------
----------- ans = min(ans, self.shortest[c] + self.path[u][h]) -------
------- return ans -------------------------------------------------
```

5. Math I Algebra

6. Math II Combinatorics

6.0.1. *Lucas Theorem.*

```
_fact_cache = [] -------------------------------------------------------
_last_p = -1 -----------------------------------------------------------
--------------------------------------------------------------------------
def lucas(n, k, p): ----------------------------------------------------
--- if k < 0 or k > n: -----------------------------------------------
------- return 0 ---------------------------------------------------
--- if k == 0 or n == k: ---------------------------------------------
------- return 1 ---------------------------------------------------
--------------------------------------------------------------------
--- global _fact_cache, _last_p --------------------------------------
--- if _last_p != p: -------------------------------------------------
------- _fact_cache = [1] * p --------------------------------------
------- for i in range(2, p): --------------------------------------
----------- _fact_cache[i] = (_fact_cache[i - 1] * i) % p ----------
------- _last_p = p ------------------------------------------------
--------------------------------------------------------------------
--- res_num = 1 ------------------------------------------------------
--- res_den = 1 ------------------------------------------------------
--------------------------------------------------------------------
--- while n > 0 or k > 0: --------------------------------------------
------- ni, ki = n % p, k % p --------------------------------------
------- if ki > ni: ------------------------------------------------
----------- return 0 ---------------------------------------------
------- res_num = (res_num * _fact_cache[ni]) % p ------------------
------- res_den = (res_den * _fact_cache[ki] * _fact_cache[ni - ki]) %
↪  p
------- n //= p ----------------------------------------------------
------- k //= p ----------------------------------------------------
--- return (res_num * pow(res_den, p - 2, p)) % p --------------------
# Input Format: print(lucas(n, k, p)) ------------------------------
```

6.0.2. *Catalan Numbers.*

```
def catalan(n): --------------------------------------------------------
--- if (n == 0 or n == 1): -------------------------------------------
------- return 1 ---------------------------------------------------
--- ------------------------------------------------------------------
--- dp = [0] * (n + 1) -----------------------------------------------
--- dp[0] = dp[1] = 1 ------------------------------------------------
--------------------------------------------------------------------
--- for i in range(2, n + 1): ----------------------------------------
------- for j in range(i): -----------------------------------------
----------- dp[i] += dp[j] * dp[i-j-1] ---------------------------
--------------------------------------------------------------------
--- return dp[n] -----------------------------------------------------
--------------------------------------------------------------------
# for i in range(10): ----------------------------------------------
#     print(catalan(i)) ------------------------------------------
```

6.0.3. *Kth Permutation.*

```
#fact = [1, 1, 2, 6, 24, 120, 720, 5040, 40320, 362880] ----------------
--------------------------------------------------------------------------
def findKthPermutation(n, k): ------------------------------------------
--- nums = [str(i) for i in range(1, n + 1)] -------------------------
--------------------------------------------------------------------
--- fact = [1] * n ---------------------------------------------------
--- for i in range(1, n): --------------------------------------------
------- fact[i] = fact[i - 1] * i ----------------------------------
--------------------------------------------------------------------
--- k -= 1 -----------------------------------------------------------
--- res = [] ---------------------------------------------------------
--------------------------------------------------------------------
--- for i in range(n - 1, -1, -1): -----------------------------------
------- idx = k // fact[i] -----------------------------------------
------- k %= fact[i] -----------------------------------------------
--------------------------------------------------------------------
------- res.append(nums.pop(idx)) ----------------------------------
--------------------------------------------------------------------
--- return ''.join(res) ----------------------------------------------
```

6.0.4. *Factoradics.*

```
def factoradic(arr): ---------------------------------------------------
--- n = len(arr) -----------------------------------------------------
--- ft = FenwickTree(n) ----------------------------------------------
--- for i in range(1, n + 1): ----------------------------------------
------- ft.add(i, 1) -----------------------------------------------
--------------------------------------------------------------------
--- res = [] ---------------------------------------------------------
--- for val in arr: --------------------------------------------------
------- bit_val = val + 1 ------------------------------------------
------- s = ft.query(bit_val - 1) ----------------------------------
------- ft.add(bit_val, -1) ----------------------------------------
------- res.append(s) ----------------------------------------------
--- return res -------------------------------------------------------
--------------------------------------------------------------------
def permute(fact, n): --------------------------------------------------
--- ft = FenwickTree(n) ----------------------------------------------
--- for i in range(1, n + 1): ----------------------------------------
------- ft.add(i, 1) -----------------------------------------------
--------------------------------------------------------------------
--- res = [] ---------------------------------------------------------
--- for val in fact: -------------------------------------------------
------- idx = ft.find_kth(val) -------------------------------------
------- ft.add(idx, -1) --------------------------------------------
------- res.append(idx - 1) ----------------------------------------
--- return res -------------------------------------------------------
```

7. Math III Number Theory

7.0.1. *Gcd Subset Counter.*

```
def gcd_subset_count(arr): ---------------------------------------------
--- n = len(arr) -----------------------------------------------------
--- mx = max(arr) ----------------------------------------------------
--------------------------------------------------------------------
--- f = [0] * (mx + 1) -----------------------------------------------
--- for x in arr: ----------------------------------------------------
------- f[x] += 1 --------------------------------------------------
--------------------------------------------------------------------
--- gcnt = [0] * (mx + 1) --------------------------------------------
--------------------------------------------------------------------
--- for i in range(mx, 0, -1): ---------------------------------------
------- add = 0 ----------------------------------------------------
--------------------------------------------------------------------
------- for j in range(i, mx + 1, i): ------------------------------
----------- add += f[j] ------------------------------------------
--------------------------------------------------------------------
------- total = (1 << add) - 1  # 2^add - 1 ------------------------
```

```
-------------------------------------------------------------------------
------- sub = 0 ---------------------------------------------------------
------- for j in range(2 * i, mx + 1, i): -------------------------------
----------- sub += gcnt[j] ------------------------------------------------
-------------------------------------------------------------------------
------- gcnt[i] = total - sub -------------------------------------------
-------------------------------------------------------------------------
--- return gcnt -------------------------------------------------------
```

7.0.2. *Linear Diophantine.*

```
def gcd(a, b): ----------------------------------------------------------
--- if a == 0: --------------------------------------------------------
------- return b ----------------------------------------------------------
--- return gcd(b % a, a) ----------------------------------------------
-------------------------------------------------------------------------
def isPossible(a, b, c): ------------------------------------------------
--- g = gcd(a, b) -----------------------------------------------------
--- -------------------------------------------------------------------
--- return c % g == 0 -------------------------------------------------
```

7.0.3. *Chinese Remainder Theorem.*

```
def gcd_extended(a, b): -------------------------------------------------
--- if a == 0: --------------------------------------------------------
------- return b, 0, 1 ----------------------------------------------------
--- gcd, x1, y1 = gcd_extended(b % a, a) ------------------------------
--- x = y1 - (b // a) * x1 --------------------------------------------
--- y = x1 ------------------------------------------------------------
--- return gcd, x, y --------------------------------------------------
-------------------------------------------------------------------------
def find_min_x(num, rem): -----------------------------------------------
--- prod = 1 ----------------------------------------------------------
--- for n in num: -----------------------------------------------------
------- prod *= n ---------------------------------------------------------
-------------------------------------------------------------------------
--- result = 0 --------------------------------------------------------
--- for i in range(len(num)): -----------------------------------------
------- prod_i = prod // num[i] -------------------------------------------
------- _, inv_i, _ = gcd_extended(prod_i, num[i]) ------------------------
------- result += rem[i] * prod_i * inv_i ---------------------------------
-------------------------------------------------------------------------
--- return result % prod ----------------------------------------------
```

7.0.4. *Euler Totient.*

```
def totient(n): ---------------------------------------------------------
--- if n <= 1: --------------------------------------------------------
------- return 1 ----------------------------------------------------------
-------------------------------------------------------------------------
--- tot = n -----------------------------------------------------------
--- i = 2 -------------------------------------------------------------
-------------------------------------------------------------------------
--- while i * i <= n: -------------------------------------------------
------- if n % i == 0: ----------------------------------------------------
----------- tot -= tot // i ---------------------------------------------
----------- while n % i == 0: -------------------------------------------
--------------- n //= i -------------------------------------------------
------- i += 1 ------------------------------------------------------------
-------------------------------------------------------------------------
--- if n > 1: ---------------------------------------------------------
------- tot -= tot // n ---------------------------------------------------
-------------------------------------------------------------------------
--- return tot --------------------------------------------------------
```

7.0.5. *Extended Euclidean Algorithm.*

```
def mod(x, m): ----------------------------------------------------------
--- if m == 0: --------------------------------------------------------
------- return 0 ----------------------------------------------------------
--- if m < 0: ---------------------------------------------------------
------- m = -m ------------------------------------------------------------
--- return (x % m + m) % m --------------------------------------------
-------------------------------------------------------------------------
def extended_euclid(a, b): ----------------------------------------------
--- if b == 0: --------------------------------------------------------
------- return a, 1, 0 ----------------------------------------------------
-------------------------------------------------------------------------
--- g, x1, y1 = extended_euclid(b, a % b) -----------------------------
-------------------------------------------------------------------------
--- x = y1 ------------------------------------------------------------
--- y = x1 - (a // b) * y1 --------------------------------------------
-------------------------------------------------------------------------
--- return g, x, y ----------------------------------------------------
```

7.0.6. *Modular Exponentiation.*

```
def mod_pow(b, e, m): ---------------------------------------------------
--- res = 1 % m -------------------------------------------------------
-------------------------------------------------------------------------
--- while e > 0: ------------------------------------------------------
------- if e & 1: ---------------------------------------------------------
----------- res = (res * b) % m -----------------------------------------
------- b = (b * b) % m ---------------------------------------------------
------- e >>= 1 -----------------------------------------------------------
-------------------------------------------------------------------------
--- return res --------------------------------------------------------
```

7.0.7. *Mod Inverse.*

```
def extended_euclid(a, b): ----------------------------------------------
--- if b == 0: --------------------------------------------------------
------- return a, 1, 0 ----------------------------------------------------
-------------------------------------------------------------------------
--- g, x1, y1 = extended_euclid(b, a % b) -----------------------------
-------------------------------------------------------------------------
--- x = y1 ------------------------------------------------------------
--- y = x1 - (a // b) * y1 --------------------------------------------
-------------------------------------------------------------------------
--- return g, x, y ----------------------------------------------------
-------------------------------------------------------------------------
-------------------------------------------------------------------------
def modinv(a, m): -------------------------------------------------------
--- g, x, y = extended_euclid(a, m) -----------------------------------
-------------------------------------------------------------------------
--- if g != 1: --------------------------------------------------------
------- return 0  # no inverse --------------------------------------------
-------------------------------------------------------------------------
--- return x % m ------------------------------------------------------
```

7.0.8. *Modulo Solver.*

```
def extended_euclid(a, b): ----------------------------------------------
--- if b == 0: --------------------------------------------------------
------- return a, 1, 0 ----------------------------------------------------
-------------------------------------------------------------------------
--- g, x1, y1 = extended_euclid(b, a % b) -----------------------------
-------------------------------------------------------------------------
--- x = y1 ------------------------------------------------------------
--- y = x1 - (a // b) * y1 --------------------------------------------
-------------------------------------------------------------------------
--- return g, x, y ----------------------------------------------------
-------------------------------------------------------------------------
-------------------------------------------------------------------------
def modsolver(a, b, m): -------------------------------------------------
--- g, x, y = extended_euclid(a, m) -----------------------------------
-------------------------------------------------------------------------
--- if b % g != 0: ----------------------------------------------------
------- return (-1, -1) ---------------------------------------------------
-------------------------------------------------------------------------
--- m_reduced = m // g ------------------------------------------------
-------------------------------------------------------------------------
--- x0 = (x * (b // g)) % m_reduced -----------------------------------
-------------------------------------------------------------------------
--- return (x0, m_reduced) --------------------------------------------
```

7.0.9. *Primitive Root.*

```
def primitive_root(m): --------------------------------------------------
--- if m == 2: --------------------------------------------------------
------- return 1 ----------------------------------------------------------
-------------------------------------------------------------------------
--- phi = m - 1 -------------------------------------------------------
-------------------------------------------------------------------------
--- factors = set() ---------------------------------------------------
--- x = phi -----------------------------------------------------------
--- i = 2 -------------------------------------------------------------
-------------------------------------------------------------------------
--- while i * i <= x: -------------------------------------------------
------- if x % i == 0: ----------------------------------------------------
----------- factors.add(i) ----------------------------------------------
----------- while x % i == 0: -------------------------------------------
--------------- x //= i -------------------------------------------------
------- i += 1 ------------------------------------------------------------
-------------------------------------------------------------------------
--- if x > 1: ---------------------------------------------------------
------- factors.add(x) ----------------------------------------------------
-------------------------------------------------------------------------
--- for g in range(2, m): ---------------------------------------------
------- ok = True ---------------------------------------------------------
------- for f in factors: -------------------------------------------------
----------- if mod_pow(g, phi // f, m) == 1: ----------------------------
--------------- ok = False ----------------------------------------------
--------------- break ---------------------------------------------------
------- if ok: ------------------------------------------------------------
----------- return g ----------------------------------------------------
-------------------------------------------------------------------------
--- return -1 ---------------------------------------------------------
```

7.0.10. *Josephus.*

```
def josephus(n, k): -----------------------------------------------------
--- res = 0 -----------------------------------------------------------
--- for i in range(2, n + 1): -----------------------------------------
------- res = (res + k) % i -----------------------------------------------
--- return res --------------------------------------------------------
```

8. Math IV Numerical Methods

8.0.1. *Fast Square Testing.*

```
import math -------------------------------------------------------------
-------------------------------------------------------------------------
class FastSquareTester: --------------------------------------------------
--- def __init__(self): -----------------------------------------------
------- self.M = 0 --------------------------------------------------------
------- self.init_is_square() ---------------------------------------------
-------------------------------------------------------------------------
--- def init_is_square(self): -----------------------------------------
------- for i in range(64): -----------------------------------------------
----------- self.M |= (1 << (i * i % 64)) -------------------------------
-------------------------------------------------------------------------
--- def is_square(self, x): -------------------------------------------
------- if x < 0: return False --------------------------------------------
------- if x == 0: return True --------------------------------------------
-------------------------------------------------------------------------
------- if not (self.M & (1 << (x & 63))): --------------------------------
----------- return False ------------------------------------------------
-------------------------------------------------------------------------
------- c = (x & -x).bit_length() - 1 -------------------------------------
------- if c & 1: ---------------------------------------------------------
----------- return False ------------------------------------------------
-------------------------------------------------------------------------
------- temp_x = x >> c ---------------------------------------------------
------- if (temp_x & 7) != 1: ---------------------------------------------
----------- return False ------------------------------------------------
-------------------------------------------------------------------------
------- r = int(math.isqrt(temp_x)) ---------------------------------------
```

```
------- return r * r == temp_x
tester = FastSquareTester()
```

8.0.2. *Simpson Integration.*

```
def f(x):
--- return x ** 2
def simpson_integration(a, b, n=1000000):
--- if n % 2 != 0:
------- n += 1

--- h = (b - a) / n
--- s = f(a) + f(b)

--- for i in range(1, n):
------- x = a + h * i
------- if i % 2 == 1:
----------- s += 4 * f(x)
------- else:
----------- s += 2 * f(x)

--- return s * (h / 3)
```

9. Strings

9.0.1. *Longest Common Prefix.*

```
def build_lcp(s, sa):
--- n = len(s)
--- rank = [0] * n
--- lcp = [0] * n

--- for i in range(n):
------- rank[sa[i]] = i

--- k = 0
--- for i in range(n):
------- if rank[i] == 0:
----------- k = 0
----------- continue

------- j = sa[rank[i] - 1]

------- while i + k < n and j + k < n and s[i + k] == s[j + k]:
----------- k += 1

------- lcp[rank[i]] = k

------- if k > 0:
----------- k -= 1

--- return lcp
```

9.0.2. *Hasher.*

```
class Hasher:
--- def __init__(self, s, bases=[31, 37], mod=10 ** 9 + 7):
------- self.bases = bases
------- self.mod = mod
------- self.n = len(s)
------- self.num_hashes = len(bases)
------- self.p_pow = [[1] * (self.n + 1) for _ in
↪  range(self.num_hashes)]
------- self.h_ans = [[0] * (self.n + 1) for _ in
↪  range(self.num_hashes)]

------- for i, b in enumerate(self.bases):
----------- for j in range(self.n):
--------------- self.p_pow[i][j + 1] = (self.p_pow[i][j] * b) %
↪  self.mod
--------------- self.h_ans[i][j + 1] = (self.h_ans[i][j] + s[j] *
↪  self.p_pow[i][j]) % self.mod

--- def get_hash(self, l, r):
------- return tuple((self.h_ans[i][r + 1] - self.h_ans[i][l]) %
↪  self.mod
------------------- for i in range(self.num_hashes))

--- def are_equal(self, l1, r1, l2, r2):
------- if (r1 - l1) != (r2 - l2): return False

------- h1 = self.get_hash(l1, r1)
------- h2 = self.get_hash(l2, r2)

------- return all((h1[i] * self.p_pow[i][l2]) % self.mod ==
------------------ (h2[i] * self.p_pow[i][l1]) % self.mod
------------------ for i in range(self.num_hashes))
```

9.0.3. *Z Algorithm.*

9.0.4. *Aho Corasick Trie.*

```
from collections import deque
import random
import string

class Node:
--- def __init__(self):
------- self.next = {}
------- self.fail = None
------- self.count = 0

--- def add(self, s):
------- node = self
------- for c in s:
----------- if c not in node.next:
--------------- node.next[c] = Node()
----------- node = node.next[c]
------- node.count += 1

--- def prepare(self):
------- root = self
------- q = deque()

------- root.fail = root

------- for child in root.next.values():
----------- child.fail = root
----------- q.append(child)

------- while q:
----------- head = q.popleft()

----------- for c, nxt in head.next.items():
--------------- f = head.fail

--------------- while f != root and c not in f.next:
------------------- f = f.fail

--------------- nxt.fail = f.next[c] if c in f.next else root

--------------- nxt.count += nxt.fail.count

--------------- q.append(nxt)

--- def search(self, s):
------- root = self
------- p = self
------- ans = 0

------- for c in s:
----------- while p != root and c not in p.next:
--------------- p = p.fail

----------- if c in p.next:
--------------- p = p.next[c]
--------------- ans += p.count

------- return ans
```

9.0.5. *Knuth Morris Pratt.*

```
def build_kmp(f):
--- n = len(f)
--- par = [0] * (n + 1)

--- par[0] = -1
--- if n > 0:
------- par[1] = 0

--- i, j = 2, 0

--- while i <= n:
------- if f[i - 1] == f[j]:
----------- j += 1
----------- par[i] = j
----------- i += 1
------- elif j > 0:
----------- j = par[j]
------- else:
----------- par[i] = 0
----------- i += 1

--- return par

def kmp(s, f):
--- par = build_kmp(f)

--- i = 0
--- j = 0
--- ans = []

--- n, m = len(s), len(f)

--- while i + j < n:
------- if s[i + j] == f[j]:
----------- j += 1
----------- if j == m:
--------------- ans.append(i)
--------------- i += j - par[j]
--------------- if j > 0:
------------------- j = par[j]
------- else:
----------- i += j - par[j]
----------- if j > 0:
--------------- j = par[j]

--- return ans
```

9.0.6. *Trie.*

```
class Node:
--- def __init__(self):
------- self.children = {}
------- self.prefixes = 0
------- self.words = 0

class Trie:
--- def __init__(self):
------- self.root = Node()

--- def insert(self, seq):
------- cur = self.root
```

```
        for i in range(len(seq) + 1):
            cur.prefixes += 1

        if i == len(seq):
            cur.words += 1
            break

        head = seq[i]

        if head not in cur.children:
            cur.children[head] = Node()

        cur = cur.children[head]

    def countMatches(self, seq):
        cur = self.root

        for c in seq:
            if c not in cur.children:
                return 0
            cur = cur.children[c]

        return cur.words

    def countPrefixes(self, seq):
        cur = self.root

        for c in seq:
            if c not in cur.children:
                return 0
            cur = cur.children[c]

        return cur.prefixes
```

9.0.7. *Suffix Array.*

```
class SuffixArray:
    def __init__(self):
        self.n = 0
        self.suffix = []
        self.equiv = []
        self.equiv_pair = []
        self.s = ""

    def build(self, s):
        if not s.endswith('$'):
            s += '$'

        self.s = s
        self.n = len(s)

        self.suffix = list(range(self.n))
        self.equiv = [0] * self.n
        self.equiv_pair = [(0, 0)] * self.n

        self.suffix.sort(key=lambda i: s[i])

        cls = 0
        for i in range(self.n):
            if i == 0 or s[self.suffix[i]] != s[self.suffix[i - 1]]:
                cls += 1
            self.equiv[self.suffix[i]] = cls

        k = 1
        while k < self.n:
            for i in range(self.n):
                self.equiv_pair[i] = (
                    self.equiv[i],
                    self.equiv[(i + k) % self.n]
                )

            self.suffix.sort(key=lambda i: self.equiv_pair[i])

            cls = 0
            new_equiv = [0] * self.n

            for i in range(self.n):
                if i == 0 or self.equiv_pair[self.suffix[i]] !=
↪   self.equiv_pair[self.suffix[i - 1]]:
                    cls += 1
                new_equiv[self.suffix[i]] = cls

            self.equiv = new_equiv
            k <<= 1

    def _cmp(self, start, pattern):
        s = self.s
        n = len(s)

        i = start
        j = 0

        while i < n and j < len(pattern):
            if s[i] != pattern[j]:
                return ord(s[i]) - ord(pattern[j])
            i += 1
            j += 1

        if j == len(pattern):
            return 0
        return -1

    def lower_bound(self, pattern):
        l, r = 0, self.n - 1
        res = self.n

        while l <= r:
            m = (l + r) // 2
            if self._cmp(self.suffix[m], pattern) >= 0:
                res = m
                r = m - 1
            else:
                l = m + 1

        return res

    def upper_bound(self, pattern):
        l, r = 0, self.n - 1
        res = self.n

        while l <= r:
            m = (l + r) // 2
            if self._cmp(self.suffix[m], pattern) > 0:
                res = m
                r = m - 1
            else:
                l = m + 1

        return res

    def count_occurrences(self, pattern):
        L = self.lower_bound(pattern)
        R = self.upper_bound(pattern)
        return max(0, R - L)
```

9.0.8. *Booth Min String Rotation.*

```
def booth(s: str) -> int:
    s = s + s
    n = len(s)

    f = [-1] * n
    k = 0

    for j in range(1, n):
        i = f[j - k - 1]

        while i != -1 and s[j] != s[k + i + 1]:
            if s[j] < s[k + i + 1]:
                k = j - i - 1
            i = f[i]

        if i == -1 and s[j] != s[k]:
            if s[j] < s[k]:
                k = j
            f[j - k] = -1
        else:
            f[j - k] = i + 1

    return k
```

10. Other Algorithms

10.0.1. *Longest Increasing Subsequence.*

```
def lis(arr):
    if not arr:
        return []

    n = len(arr)
    seq = []
    back = [0] * n

    for i in range(n):
        lo, hi = 0, len(seq)
        while lo < hi:
            mid = (lo + hi) // 2
            if arr[seq[mid]] < arr[i]:
                lo = mid + 1
            else:
                hi = mid

        res = lo
        if res < len(seq):
            seq[res] = i
        else:
            seq.append(i)
        back[i] = -1 if res == 0 else seq[res - 1]

    ans = []
    at = seq[-1]
    while at != -1:
        ans.append(at)
        at = back[at]

    return ans[::-1]
```

10.0.2. *Cycle Find.*

```
def find_cycle(x0, f):
    t = f(x0)
    h = f(f(x0))

    while t != h:
        t = f(t)
        h = f(f(h))

    mu = 0
    h = x0
    while t != h:
        t = f(t)
        h = f(h)
        mu += 1

    lam = 1
```

```
    h = f(t)
    while t != h:
        h = f(h)
        lam += 1

    return mu, lam
```

10.0.3. *2Sat.*

```
class TwoSAT:
    def __init__(self, n):
        self.n = n
        self.N = 2 * n
        self.adj = [[] for _ in range(self.N)]

    def var(self, x):
        return x << 1

    def neg(self, x):
        return x ^ 1

    def add_imp(self, x, y):
        self.adj[x].append(y)

    def add_or(self, x, y):
        self.add_imp(self.neg(x), y)
        self.add_imp(self.neg(y), x)

    def sat(self):
        n = self.N
        adj = self.adj

        index = 0
        stack = []
        onstack = [False] * n
        ids = [-1] * n
        low = [0] * n
        comp = [-1] * n
        self.time = 0
        self.comp_id = 0

        def dfs(u):
            nonlocal index

            ids[u] = low[u] = index
            index += 1
            stack.append(u)
            onstack[u] = True

            for v in adj[u]:
                if ids[v] == -1:
                    dfs(v)
                    low[u] = min(low[u], low[v])
                elif onstack[v]:
                    low[u] = min(low[u], ids[v])

            if low[u] == ids[u]:
                while True:
                    v = stack.pop()
                    onstack[v] = False
                    comp[v] = self.comp_id
                    if v == u:
                        break
                self.comp_id += 1

        for i in range(n):
            if ids[i] == -1:
                dfs(i)

        for i in range(0, n, 2):
            if comp[i] == comp[i ^ 1]:
                return False

        return True
```

10.0.4. *Dpll.*

```
class SAT:
    def __init__(self):
        self.clauses = []
        self.n = 0

    def var(self):
        self.n += 1
        return self.n

    def clause(self, c):
        self.clauses.append(c)

    def clause_sat(self, clause, assign):
        unknown = False

        for x in clause:
            v = abs(x)
            val = assign[v]

            if x > 0 and val == 1:
                return True
            if x < 0 and val == 0:
                return True

            if val == -1:
                unknown = True

        return False if not unknown else None

    def propagate(self, assign):
        changed = True

        while changed:
            changed = False

            for clause in self.clauses:
                sat = False
                unassigned = []

                for x in clause:
                    v = abs(x)

                    if x > 0 and assign[v] == 1:
                        sat = True
                        break
                    if x < 0 and assign[v] == 0:
                        sat = True
                        break

                    if assign[v] == -1:
                        unassigned.append(x)

                if sat:
                    continue

                if len(unassigned) == 0:
                    return False  # contradiction

                if len(unassigned) == 1:
                    x = unassigned[0]
                    v = abs(x)
                    val = 1 if x > 0 else 0

                    if assign[v] != -1 and assign[v] != val:
                        return False

                    if assign[v] == -1:
                        assign[v] = val
                        changed = True

        return True

    def solve_dfs(self, assign, idx=1):
        if idx > self.n:
            return True

        if assign[idx] != -1:
            return self.solve_dfs(assign, idx + 1)

        for val in [0, 1]:
            new_assign = assign[:]
            new_assign[idx] = val

            if self.propagate(new_assign):
                if self.solve_dfs(new_assign, idx + 1):
                    return True

        return False

    def solve(self):
        assign = [-1] * (self.n + 1)

        if not self.propagate(assign):
            return False

        return self.solve_dfs(assign)

    def get_value(self, x):
        return self.assign[x] == 1
```

10.0.5. *Stable Marriage.*

```
from collections import deque

def stable_marriage(n, m_pref, w_pref):
    w_rank = [[0] * n for _ in range(n)]

    for w in range(n):
        for i, man in enumerate(w_pref[w]):
            w_rank[w][man] = i

    q = deque(range(n))

    at = [0] * n

    eng = [-1] * n

    res = [-1] * n

    while q:
        m = q.popleft()

        while at[m] < n:
            w = m_pref[m][at[m]]
            at[m] += 1

            if eng[w] == -1:
                eng[w] = m
                res[m] = w
                break

            if w_rank[w][m] < w_rank[w][eng[w]]:
                q.append(eng[w])
                eng[w] = m
                res[m] = w
                break

    return res
```

10.0.6. *Dates.*

```
def c_div(a, b):
    return int(a / b)

def date_to_int(y, m, d):
    k = c_div(m - 14, 12)

    return (
        1461 * (y + 4800 + k) // 4
        + 367 * (m - 2 - k * 12) // 12
        - 3 * ((y + 4900 + k) // 100) // 4
        + d - 32075
    )

def int_to_day(jd):
    return jd % 7

def int_to_date(jd):
    x = jd + 68569
    n = 4 * x // 146097
    x -= (146097 * n + 3) // 4

    i = (4000 * (x + 1)) // 1461001
    x -= 1461 * i // 4 - 31

    j = 80 * x // 2447
    d = x - 2447 * j // 80

    x = j // 11
    m = j + 2 - 12 * x
    y = 100 * (n - 49) + i + x

    return y, m, d
```

10.0.7. *Simulated Annealing.*

```
import random
import math
import time

def simulated_annealing(n, seconds):
    rng = random.Random()

    sol = list(range(1, n + 1))
    rng.shuffle(sol)

    def score(arr):
        return sum(abs(arr[i] - arr[i - 1]) for i in range(1, n))

    cur = score(sol)

    T0 = 100.0
    T1 = 0.001
    start = time.time()
    iters = 0

    while True:
        if iters % 16 == 0:
            progress = (time.time() - start) / seconds
            if progress >= 1.0:
                break
            temp = T0 * (T1 / T0) ** progress

        a = rng.randint(0, n - 2)

        x, y = sol[a], sol[a + 1]

        delta = 0

        if a > 0:
            delta += abs(sol[a - 1] - y) - abs(sol[a - 1] - x)

        if a + 2 < n:
            delta += abs(x - sol[a + 2]) - abs(y - sol[a + 2])

        accept = delta <= 0 or rng.random() < math.exp(-delta / temp)

        if accept:
            sol[a], sol[a + 1] = sol[a + 1], sol[a]
            cur += delta

        iters += 1

    return cur, sol
```

10.0.8. *Fast Input Reading.*

```
import sys

data = sys.stdin.buffer.read().split()
it = iter(data)

def next_int():
    return int(next(it))
```

10.0.9. *Bit Hacks.*

```
def snoob(x):
    y = x & -x
    z = x + y
    return z | (((x ^ z) // y) >> 2)
```